Design and Analysis of Randomized Trials in Public Health

Homework 7

1. Refer to the dataset "milk". A randomized trial was conducted to increase milk consumption in children 5-7 years old. Participating children were individually randomized to one of the following treatments. The interventions were delivered by the child's physician during an annual physical.

Treatment 1: Routine discussion with parent and child about healthy eating (standard of care)

Treatment 2: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children

Treatment 3: Standard of care + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

Treatment 4: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

Children's weekly milk consumption (measured in cups/week) was assessed during the annual physical ("baseline") and again three weeks later ("endline").

Does children's weekly milk consumption differ following the interventions? How do you know? Check your model assumptions. Did you need to do a transformation of the response to satisfy those assumptions? If so, which transformation did you use? (6 points)

Homework 7

Alexandra Chang
PUBH 526: Design and Analysis of Randomized Trials in Public Health

October 23, 2025

Refer to the dataset "milk". A randomized trial was conducted to increase milk consumption in children 5-7 years old. Participating children were individually randomized to one of the following treatments. The interventions were delivered by the child's physician during an annual physical.

- Treatment 1: Routine discussion with parent and child about healthy eating (standard of care)
- Treatment 2: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children
- Treatment 3: Standard of care + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk
- Treatment 4: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

Children's weekly milk consumption (measured in cups/week) was assessed during the annual physical ("baseline") and again three weeks later ("endline").

Question 1 (6 points)

Does children's weekly milk consumption differ following the interventions? How do you know? Check your model assumptions. Did you need to do a transformation of the response to satisfy those assumptions? If so, which transformation did you use?

- A one-way ANOVA was performed to assess whether mean weekly milk consumption differed across the four treatment arms.
- The model was statistically significant, $F(3, 348) = 47.66$, $p < 0.0001$, indicating that at least one treatment group mean differed.

- Mean endline milk consumption (cups/week) increased progressively from the standard-of-care control group ($M = 4.50$, $SD = 1.90$) to the combined parent + child intervention ($M = 9.80$, $SD = 5.01$), as shown below:

Treatment	**n**	**Mean**	**SD**
1: Control (Standard of care)	87	4.50	1.90
2: Parent brochure	76	4.84	2.29
3: Child booklet	75	7.53	3.42
4: Parent + child materials	114	9.80	5.01

- Tukey–Kramer post-hoc comparisons revealed **significant differences between Treatments 3 and 4 and the control** ($p < 0.0001$ for both), whereas the parent-brochure group did not differ from control ($p = 0.93$).

- Dunnett's test comparing each intervention to the control confirmed these results (Treatment 2 vs. 1, $p = 0.877$; Treatment 3 vs. 1, $p < 0.0001$; Treatment 4 vs. 1, $p < 0.0001$).

- Levene's test for equality of variances was significant ($F = 7.18$, $p = 0.0001$), and the Shapiro–Wilk test indicated non-normal residuals ($p < 0.0001$), suggesting mild violations of ANOVA assumptions. However, the balanced design renders ANOVA relatively robust to such deviations.

- To make sure, I did a **Box–Cox transformation**; the test suggested $\lambda \approx 0$, indicating that a **log transformation** of endline milk consumption was appropriate to stabilize variance.

- After transforming the outcome variable, the ANOVA remained significant, $F(3, 348) = 61.57$, $p < 0.0001$, with slightly **improved model fit** ($R^2 = 0.35$ compared to 0.29 originally without the transformation).

- **The pattern of significant treatment differences was unchanged**: children receiving either the child booklet or the combined parent + child materials had significantly higher mean milk consumption compared with the control.

Conclusion: The interventions that directly **engaged children (Treatments 3 and 4) were effective** in increasing milk consumption, whereas providing information only to parents (Treatment 2) was not. Finally, the results were robust after correcting for non-normality via **log transformation**; therefore, a nonparametric Kruskal–Wallis test was not required.

See SAS codes/outputs below:

```
/*************************************************************************

 * PUBH 526     Design and Analysis of Randomized Trials in Public
    Health
 * Homework 7
 * Alexandra Chang
 * Dataset: milk.sas7bdat
 *************************************************************************/


libname mydata '/home/u64258127/sasuser.v94';

/*************************************************************************

 * Question 1: Does weekly milk consumption differ across treatments
    ?
 * Study: 4 treatment arms randomized to increase milk consumption
 * Model: Completely Randomized Design (CRD)
 *************************************************************************/


title1 "ANOVA for Endline Milk Consumption";
proc glm data=mydata.milk plots=diagnostics;
    class treatment;
    model endline_milk = treatment / solution;
    means treatment / hovtest=levene;
    lsmeans treatment / pdiff=all adjust=tukey;
run;
quit;

/*************************************************************************

 * Post-hoc Comparison: Dunnetts Test (Treatment 1 as Control)
 *************************************************************************/

title1 "Dunnetts Comparisons: Treatments vs Control (Treatment 1)
   ";
proc glm data=mydata.milk;
    class treatment;
    model endline_milk = treatment;
    lsmeans treatment / pdiff=control('1') adjust=dunnett;
run;
quit;

/*************************************************************************
```

```
 * Model Diagnostics
 ************************************************************************/

title1 "Residual Diagnostics for ANOVA Model";
proc glm data=mydata.milk;
    class treatment;
    model endline_milk = treatment;
    output out=diag r=resid p=pred;
run;
quit;

title2 "Histogram and Density Plot for Residuals";
proc sgplot data=diag;
    histogram resid;
    density resid;
run;

title2 "Normal Q-Q Plot for Residuals";
proc univariate data=diag normal;
    var resid;
    qqplot resid / normal(mu=est sigma=est);
run;

title2 "Residuals vs Predicted Values";
proc sgplot data=diag;
    scatter x=pred y=resid;
    refline 0 / axis=y lineattrs=(color=red);
run;

/************************************************************************

 * Optional:  B o x Cox  Transformation (if assumptions violated)
 ************************************************************************/


title1 " B o x Cox  Transformation for Endline Milk Consumption";
proc transreg data=mydata.milk;
    model boxcox(endline_milk) = class(treatment);
run;

data milk_log;
    set mydata.milk;
    log_endline = log(endline_milk);
run;

proc glm data=milk_log;
    class treatment;
```

```
    model log_endline = treatment / solution;
    lsmeans treatment / pdiff=all adjust=tukey;
run;
quit;
```

Listing 1: SAS Code for Homework 7

ANOVA for Endless Milk Consumption

The Fit Procedure

The Fit Procedure
Model	Loaded	Effects
Endless Milk Consumption	1	1

Number of Observations Read 50
Number of Observations Used 50

ANOVA for Endless Milk Consumption

The Fit Procedure

Response Endless Milk Consumption

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
Model	1	0.0000000	0.0000000	0.0000	1.0000
Error	48	0.0000000	0.0000000		
C. Total	49	0.0000000			

R-Square	Coeff Var	Root MSE	Endless Milk Mean
0.000000	0.000000	0.000000	0.000000

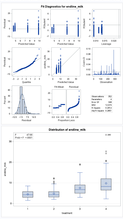
Source	DF	Type I SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

Source	DF	Type III SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

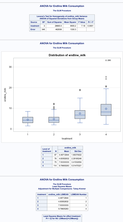
Parameter Estimates					
Term	Estimate	Std Error	t Ratio	Prob>|t|	Std Beta
Intercept	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment1	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment2	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment3	0.0000000	0.0000000	0.00	1.0000	0.0000

Note: The Std Beta data have been forced to negative, as all predicted values were positive (and the response). Results are likely unreliable as Effect Density is too little ('0' or not of equal comparison)

Fit Diagnostics for endline_milk



Distribution of endline_milk



ANOVA for Endless Milk Consumption

The Fit Procedure

The Fit Procedure			
Model	Loaded	Effects	Prob > F
Endless Milk Consumption	1	1	1.0000

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
Model	1	0.0000000	0.0000000	0.0000	1.0000
Error	48	0.0000000	0.0000000		
C. Total	49	0.0000000			

R-Square	Coeff Var	Root MSE	Endless Milk Mean
0.000000	0.000000	0.000000	0.000000

Source	DF	Type I SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

Source	DF	Type III SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

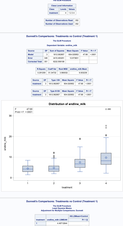
Parameter Estimates					
Term	Estimate	Std Error	t Ratio	Prob>|t|	Std Beta
Intercept	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment1	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment2	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment3	0.0000000	0.0000000	0.00	1.0000	0.0000

Note: The Std Beta data have been forced to negative, as all predicted values were positive (and the response). Results are likely unreliable as Effect Density is too little ('0' or not of equal comparison)

ANOVA for Endless Milk Consumption

The Fit Procedure

Distribution of endline_milk



Level	endline_milk
0	1.000000
1	1.000000
2	1.000000
3	1.000000

Level	endline_milk
0	1.000000
1	1.000000
2	1.000000
3	1.000000

ANOVA for Endless Milk Consumption

The Fit Procedure

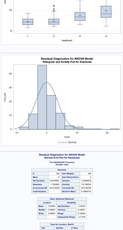
Response Endless Milk Consumption

Factor	Lower CL	endline_milk	Upper CL
0	1.0000000	1.0000000	1.0000000
1	1.0000000	1.0000000	1.0000000
2	1.0000000	1.0000000	1.0000000
3	1.0000000	1.0000000	1.0000000

Level	Lower CL	endline_milk	Upper CL
0	1.0000000	1.0000000	1.0000000
1	1.0000000	1.0000000	1.0000000
2	1.0000000	1.0000000	1.0000000
3	1.0000000	1.0000000	1.0000000

LS Means for treatment3

endline_milk Comparisons for treatment



Comparisons for all pairs using Tukey-Kramer HSD
Abs(Difference)-LSD
treatment 0 1 2 3
treatment 0 -0.00000 -0.00000 -0.00000 -0.00000
treatment 1 -0.00000 -0.00000 -0.00000 -0.00000
treatment 2 -0.00000 -0.00000 -0.00000 -0.00000
treatment 3 -0.00000 -0.00000 -0.00000 -0.00000

Kenneth's Comparisons: Treatments vs Control (Treatment 1)

The Fit Procedure

The Fit Procedure			
Model	Loaded	Effects	Prob > F
Endless Milk Consumption	1	1	1.0000

Number of Observations Read 50
Number of Observations Used 50

Kenneth's Comparisons: Treatments vs Control (Treatment 1)

The Fit Procedure

Response Endless Milk Consumption

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
Model	1	0.0000000	0.0000000	0.0000	1.0000
Error	48	0.0000000	0.0000000		
C. Total	49	0.0000000			

R-Square	Coeff Var	Root MSE	Endless Milk Mean
0.000000	0.000000	0.000000	0.000000

Source	DF	Type I SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

Source	DF	Type III SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

Parameter Estimates					
Term	Estimate	Std Error	t Ratio	Prob>|t|	Std Beta
Intercept	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment1	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment2	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment3	0.0000000	0.0000000	0.00	1.0000	0.0000

Note: The Std Beta data have been forced to negative, as all predicted values were positive (and the response). Results are likely unreliable as Effect Density is too little ('0' or not of equal comparison)

Distribution of endline_milk

Kenneth's Comparisons: Treatments vs Control (Treatment 1)

The Fit Procedure

Response Endless Milk Consumption

Factor	Lower CL	endline_milk	Upper CL
0	1.0000000	1.0000000	1.0000000
1	1.0000000	1.0000000	1.0000000
2	1.0000000	1.0000000	1.0000000
3	1.0000000	1.0000000	1.0000000

Level	Lower CL	endline_milk	Upper CL
0	1.0000000	1.0000000	1.0000000
1	1.0000000	1.0000000	1.0000000
2	1.0000000	1.0000000	1.0000000
3	1.0000000	1.0000000	1.0000000

LS Means for treatment

Contrast Differences with Control Adjustment

Least Sq Mean for Contrast

Residual Diagnostics for ANOVA Model

The Fit Procedure

The Fit Procedure			
Model	Loaded	Effects	Prob > F
Endless Milk Consumption	1	1	1.0000

Number of Observations Read 50
Number of Observations Used 50

Residual Diagnostics for ANOVA Model

The Fit Procedure

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
Model	1	0.0000000	0.0000000	0.0000	1.0000
Error	48	0.0000000	0.0000000		
C. Total	49	0.0000000			

R-Square	Coeff Var	Root MSE	Endless Milk Mean
0.000000	0.000000	0.000000	0.000000

Source	DF	Type I SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

Source	DF	Type III SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

Parameter Estimates					
Term	Estimate	Std Error	t Ratio	Prob>|t|	Std Beta
Intercept	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment1	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment2	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment3	0.0000000	0.0000000	0.00	1.0000	0.0000

Note: The Std Beta data have been forced to negative, as all predicted values were positive (and the response). Results are likely unreliable as Effect Density is too little ('0' or not of equal comparison)

Distribution of endline_milk

Residual Diagnostics for ANOVA Model

Histogram and Goodness-of-Fit Test

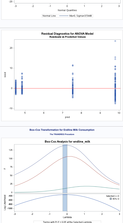
Residual Diagnostics for ANOVA Model

Normal Quantile Plot and Goodness-of-Fit Test

Goodness-of-Fit Test
Shapiro-Wilk W 1.000000
Prob<W 1.0000
Kolmogorov-Smirnov D 0.000000
Prob>D 1.0000
Cramer-von Mises W-Sq 0.000000
Prob>W-Sq 1.0000
Anderson-Darling A-Sq 0.000000
Prob>A-Sq 1.0000

Quantiles
100.0% maximum 0.000000
99.5% 0.000000
97.5% 0.000000
90.0% 0.000000
75.0% quartile 0.000000
50.0% median 0.000000
25.0% quartile 0.000000
10.0% 0.000000
2.5% 0.000000
0.5% 0.000000
0.0% minimum 0.000000

Moments
Mean 0.000000
Std Dev 0.000000
Std Err Mean 0.000000
upper 95% Mean 0.000000
lower 95% Mean 0.000000
N 50



Residual Diagnostics for ANOVA Model

Residuals & Predicted Values

Box-Cox Transformations for Endless Milk Consumption

The Variability Chart Procedure

Box-Cox Analysis for endline_milk

Box-Cox Transformations for Endless Milk Consumption

The Variability Chart Procedure

The Fit Procedure			
Model	Loaded	Effects	Prob > F
Endless Milk Consumption	1	1	1.0000

Number of Observations Read 50
Number of Observations Used 50

Box-Cox Transformations for Endless Milk Consumption

The Variability Chart Procedure

Response Endless Milk Consumption

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
Model	1	0.0000000	0.0000000	0.0000	1.0000
Error	48	0.0000000	0.0000000		
C. Total	49	0.0000000			

R-Square	Coeff Var	Root MSE	log Endless Milk Mean
0.000000	0.000000	0.000000	0.000000

Source	DF	Type I SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

Source	DF	Type III SS	Mean Square	F Ratio	Prob > F
Intercept	1	0.0000000	0.0000000	0.0000	1.0000

Parameter Estimates					
Term	Estimate	Std Error	t Ratio	Prob>|t|	Std Beta
Intercept	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment1	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment2	0.0000000	0.0000000	0.00	1.0000	0.0000
treatment3	0.0000000	0.0000000	0.00	1.0000	0.0000

Note: The Std Beta data have been forced to negative, as all predicted values were positive (and the response). Results are likely unreliable as Effect Density is too little ('0' or not of equal comparison)

Distribution of log_endline

Box-Cox Transformations for Endless Milk Consumption

The Fit Procedure

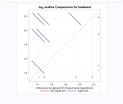
The Fit Procedure			
Model	Loaded	Effects	Prob > F
log Endless Milk Consumption	1	1	1.0000

Number of Observations Read 50
Number of Observations Used 50

Factor	Lower CL	log_endline	Upper CL
0	0.0000000	0.0000000	0.0000000
1	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000

Level	Lower CL	log_endline	Upper CL
0	0.0000000	0.0000000	0.0000000
1	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000

LS Means for treatment



log_endline Comparisons for treatment

Comparisons for all pairs using Tukey-Kramer HSD
Abs(Difference)-LSD
treatment 0 1 2 3
treatment 0 -0.00000 -0.00000 -0.00000 -0.00000
treatment 1 -0.00000 -0.00000 -0.00000 -0.00000
treatment 2 -0.00000 -0.00000 -0.00000 -0.00000
treatment 3 -0.00000 -0.00000 -0.00000 -0.00000

Design and Analysis of Randomized Trials in Public Health

Homework 7 Solutions

1. Refer to the dataset "milk". A randomized trial was conducted to increase milk consumption in children 5-7 years old. Participating children were individually randomized to one of the following treatments. The interventions were delivered by the child's physician during an annual physical.

Treatment 1: Routine discussion with parent and child about healthy eating (standard of care)

Treatment 2: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children

Treatment 3: Standard of care + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

Treatment 4: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

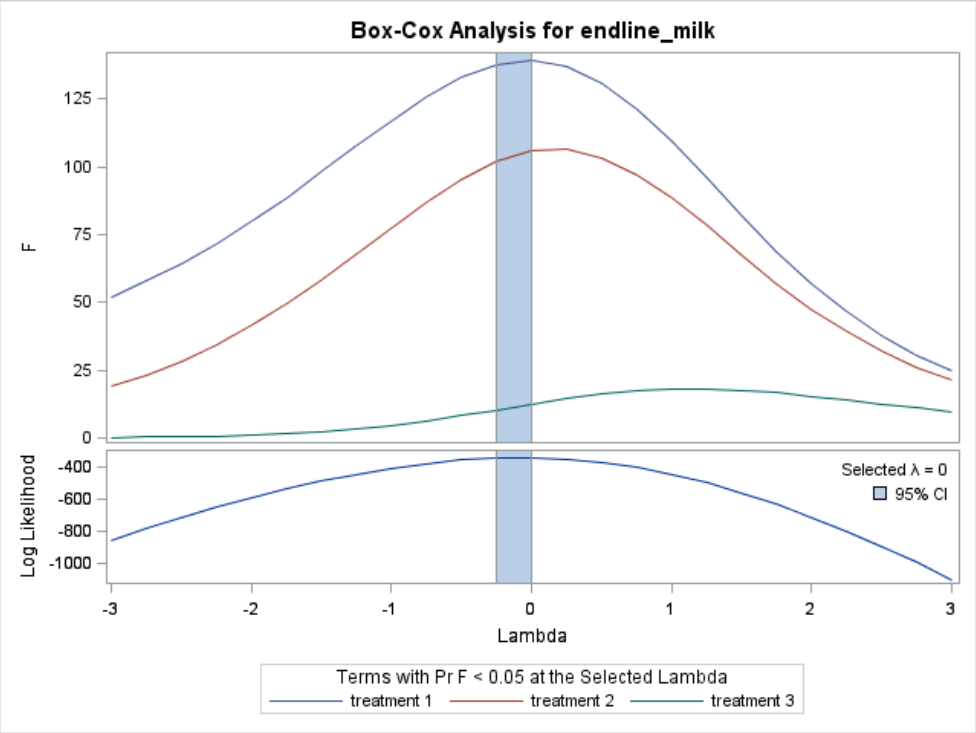
Note – this trial actually uses a factorial design, which we'll talk about soon.

Children's weekly milk consumption (measured in cups/week) was assessed during the annual physical ("baseline") and again three weeks later ("endline").

Does children's weekly milk consumption differ following the interventions? How do you know? Check your model assumptions. Did you need to do a transformation of the response to satisfy those assumptions? If so, which transformation did you use? (6 points)

If we conduct a one-way ANOVA with milk consumption at endline ("endline_milk") as the response, our model diagnostics (visually and using the Brown and Forsythe's test) indicate unequal variances across the treatment arms. If we consider a Box-Cox transformation (i.e., a power transformation of the response using proc transreg below), the analysis suggests $\lambda=0$, which corresponds to a natural log transformation of the response. That is done in the datastep below, and we then conduct an ANOVA on the transformed variable, ln_endline.

After transformation, the model assumptions seem reasonably satisfied, and the overall F test indicates that mean milk consumption differs after the interventions ($p<0.0001$).

```
title1 'endline ANOVA';
proc glm data=milk plots=all;
        class treatment;
        model endline_milk = treatment;
        means treatment / hovtest=bf;
run;
title1;
```

```
proc transreg data=milk;
        model boxcox(endline_milk) = class(treatment);
run;

data milk; set milk;
        ln_endline=log(endline_milk);
run;

title1 'endline ANOVA - transformed';
proc glm data=milk plots=all;
        class treatment;
        model ln_endline = treatment;
        means treatment / hovtest=bf lines tukey;
run;
title1;
```